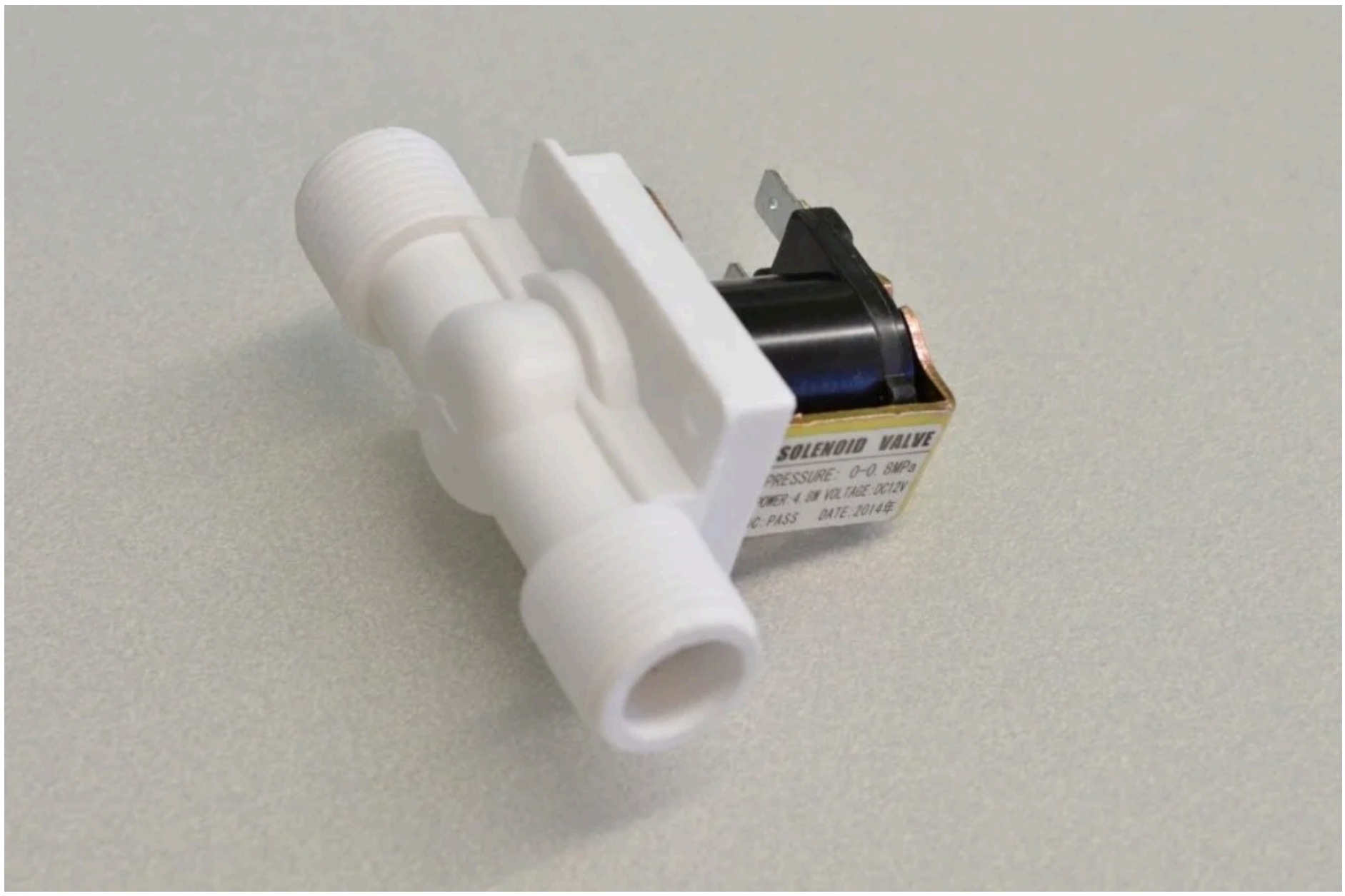
Controlling A Solenoid Valve With Arduino



PRODUCT TUTORIAL

Chris @ BCR July 2, 2015 12:01 am 187 Comments

In this tutorial we will be controlling a solenoid with an Arduino and a transistor. The solenoid we have picked for this tutorial is our Plastic Water Solenoid Valve (perfect for controlling flow to a drip irrigation system) but this tutorial can be applied to most inductive loads including relays, solenoids, and basic DC motors.

A Few Considerations:

Before choosing this solenoid valve for a project there are a few things that should be considered:

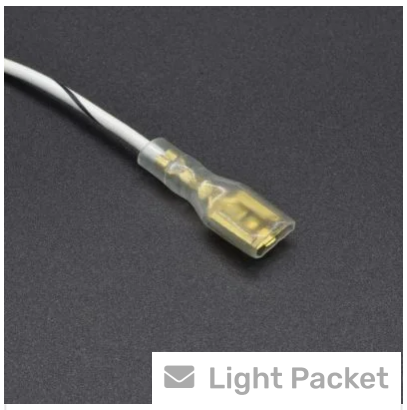
- Water can only flow in one direction through this valve.
- There is a 3 PSI minimum pressure requirement on the inlet otherwise the valve will not shut off.
- This solenoid valve is not rated for food safety or use with anything but water.

How It Works:

This valve is very similar to those found in a lawn irrigation system – the only real difference being the size. The inlet water pressure actually holds the valve closed so if you do not have any inlet water pressure (3psi minimum) the valve will never close!

The Parts Needed:

We have used the following parts in the tutorial:



Light Packet

Quick Connect Protective Cover – 1/4″ (10 Pack)

$1.00



Sale

Water Solenoid Valve – DN15

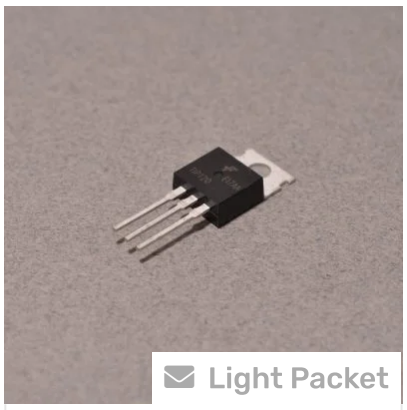
~~$10.95~~ **$8.00**



Light Packet

Quick Connect – 1/4″

$0.50



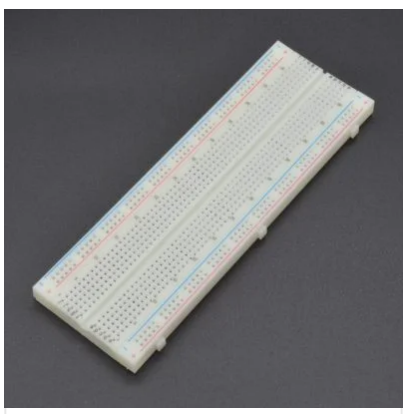
TIP120 Darlington Transistor

$2.25



Premium Male/Male Jumper Wires – 6″

$4.95



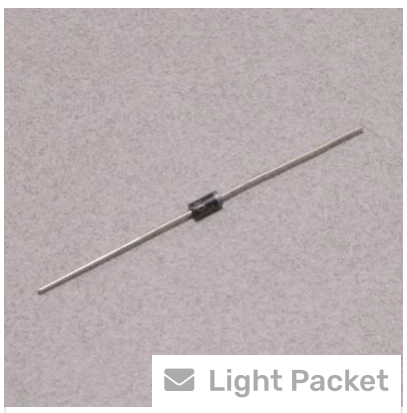
Solderless Breadboard – Large

$6.95



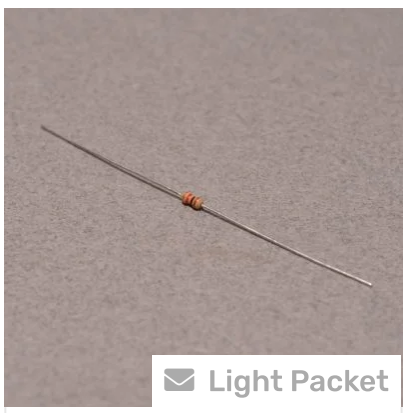
Arduino Uno R3

$44.95



1N4001 Diode – 1A 50V

$0.25



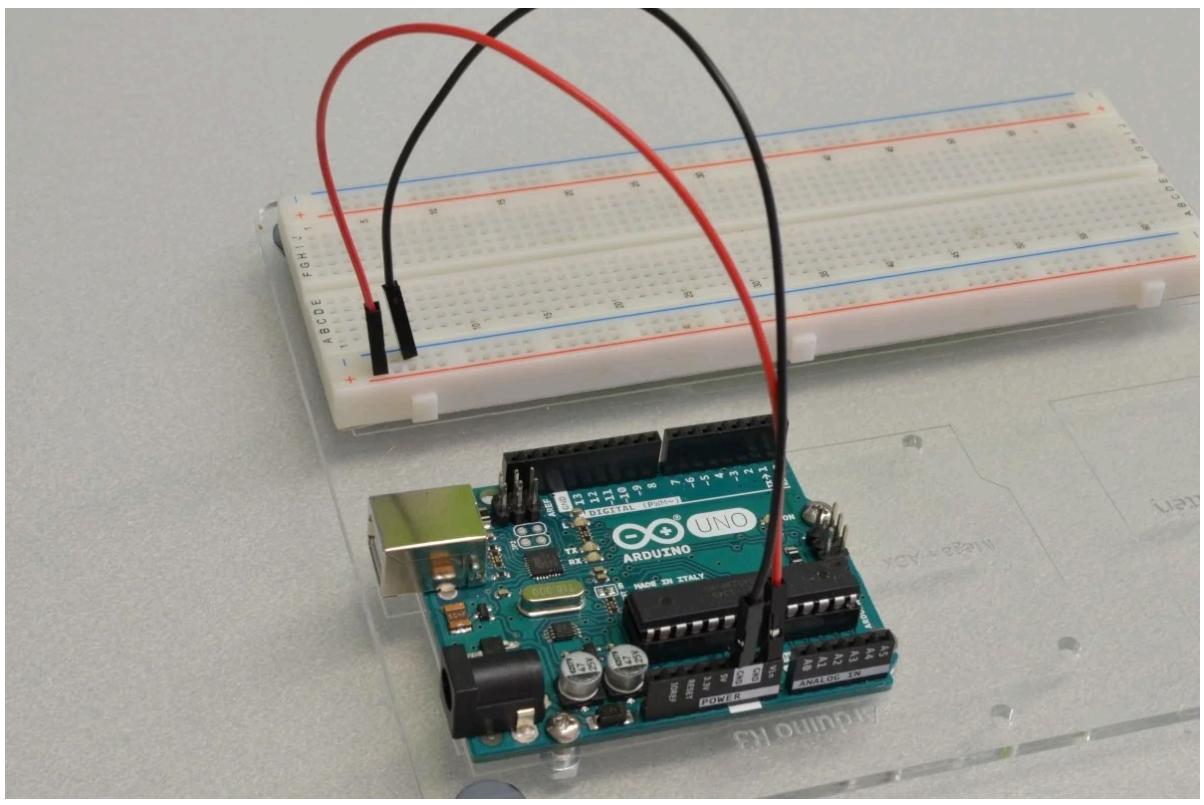
1K Ohm Resistor – 1/6W

$0.25

The Schematic

This handy little diagram shows how we will be connecting everything. Don’t worry if it looks a little overwhelming, we will be going through this step by step!

Step 1 - Powering The Breadboard



Let’s go ahead and set up the Arduino Uno and the breadboard right next to one another. The solenoid works with anywhere between 6-12V which is too high to use with the standard Arduino 5V. To get around this problem we will be using a 9V power supply – the solenoid will operate at 9V while the Arduino’s built in voltage regulator will turn that 9V into the 5V that it needs to operate. To gain access to the raw voltage going into the DC barrel jack on the Arduino Uno we will use the “Vin” pin located next to the ground pin on the Arduino.

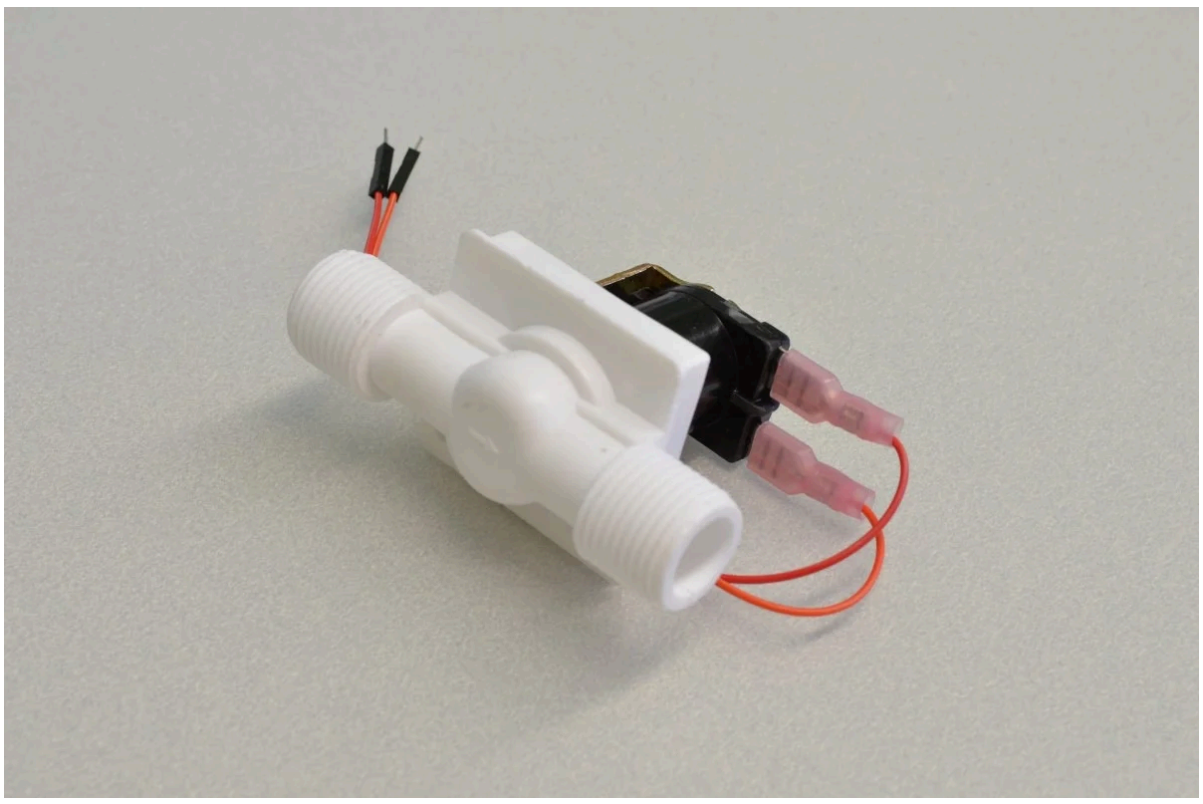
> **Please Note:**
>
> Do Not use this method of wiring with input voltages greater than 12VDC. An external regulator or separate power supply is required.

Start by connecting one of the jumper wires to the “Vin” pin on the Arduino and running it over to the positive rail on the side of the solderless breadboard. Next, run a wire from the Ground pin on the Arduino over to the negative rail on the solderless breadboard.

We now have 9VDC power on the breadboard! With the exception of the “Vin” pin, 9V is more than enough to damage your Arduino so do not plug ANY other pins from the Arduino into the positive rail on the breadboard.

6.6%

Step 2 - The Solenoid's Wire Harness

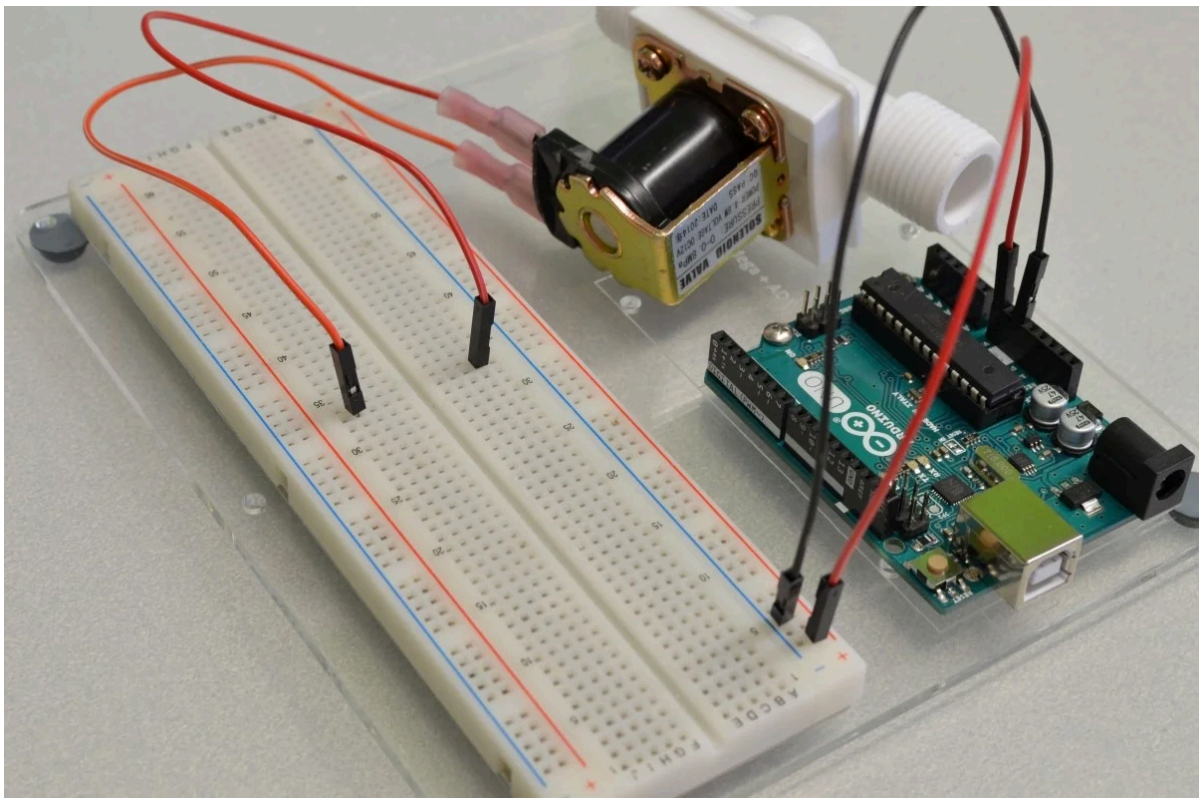


This solenoid does not have a wire harness and instead relies on 0.250″ Quick Connects. These are the best way to connect the solenoid. If you do not have Quick Connects laying around, Alligator Clips or even soldering wires to the tabs will work!

The connections on the solenoid do not matter, the coil does not care which side is positive or negative.

13.3%

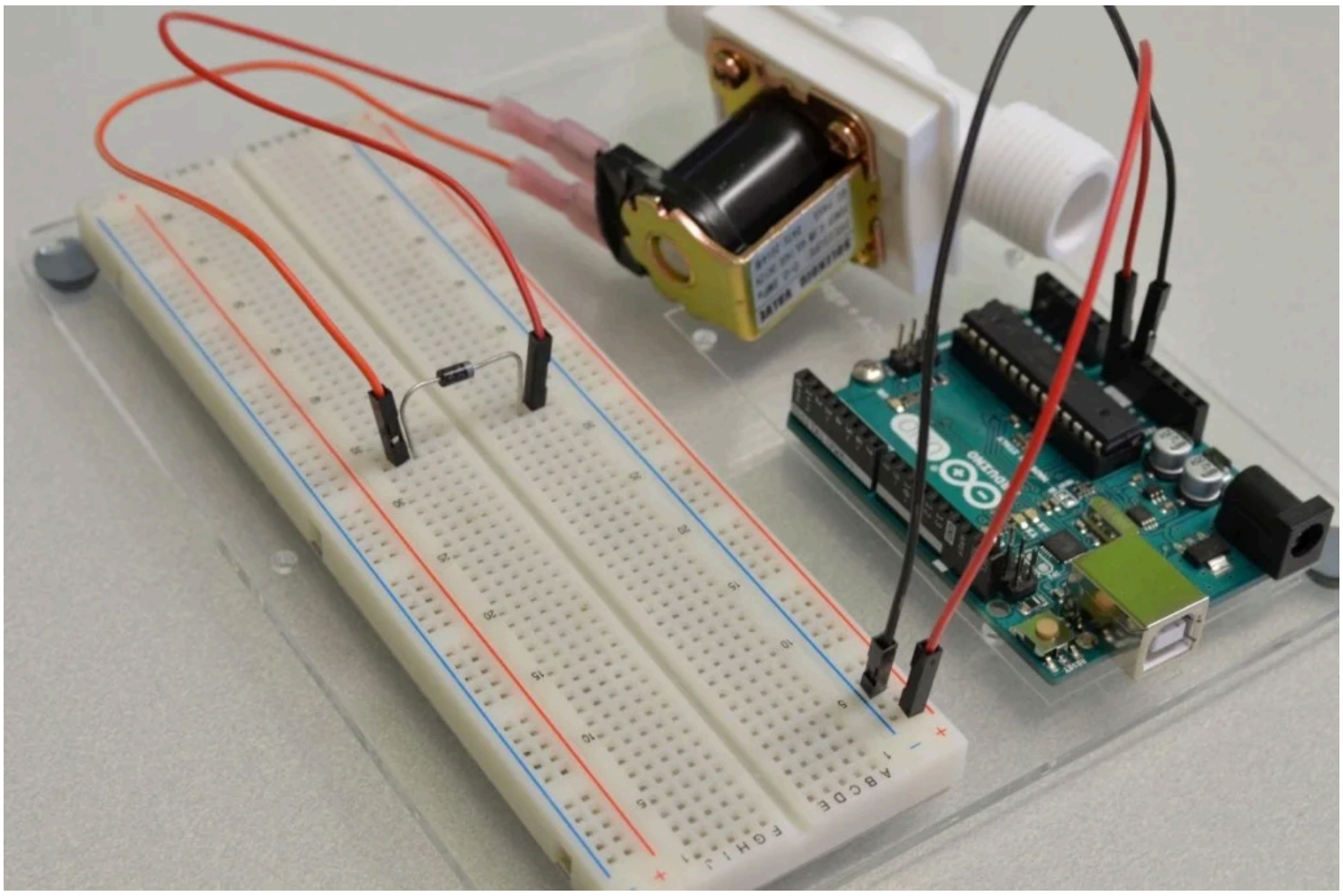
Step 3 - Solenoid To Breadboard



Connect the solenoid to the breadboard – we will need to add a diode between the two contacts so we will leave some space for that.

20%

Step 4 - Snubber Diode

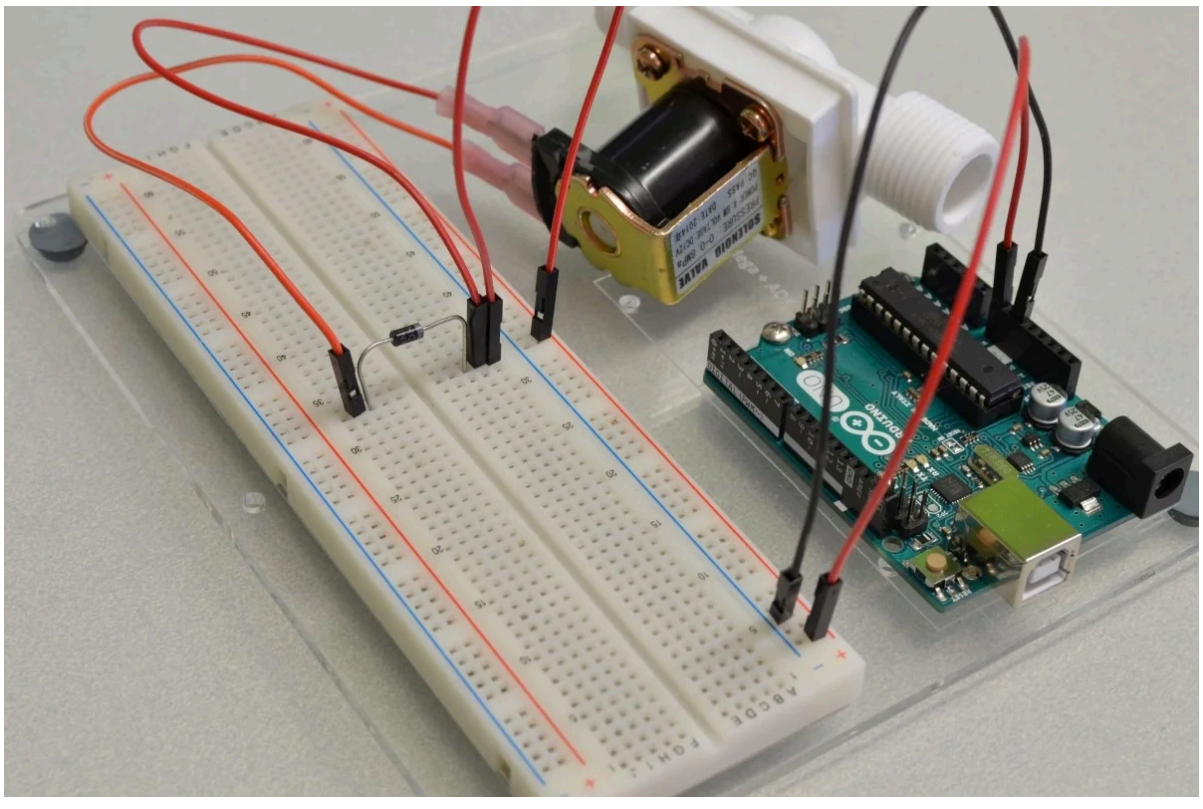


Since a solenoid is an inductive load we need to include a snubber diode across the contacts. Snubber diodes help eliminate transient voltages caused when a magnetic coil (such as those found in a motor, relay, or solenoid) suddenly loses power. Without this diode in place the transient voltage spikes can damage other elements of the circuit.

The snubber is placed from the negative side of the coil to the positive side. Since diodes only allow current to flow in one direction we need to make sure we get this right, otherwise it will be a dead short between power and ground. Ensure the side with the White stripe is connected to power/positive side of the solenoid! In our circuit the Red wire is the positive 9V so we will connect it to this side.

26.6%

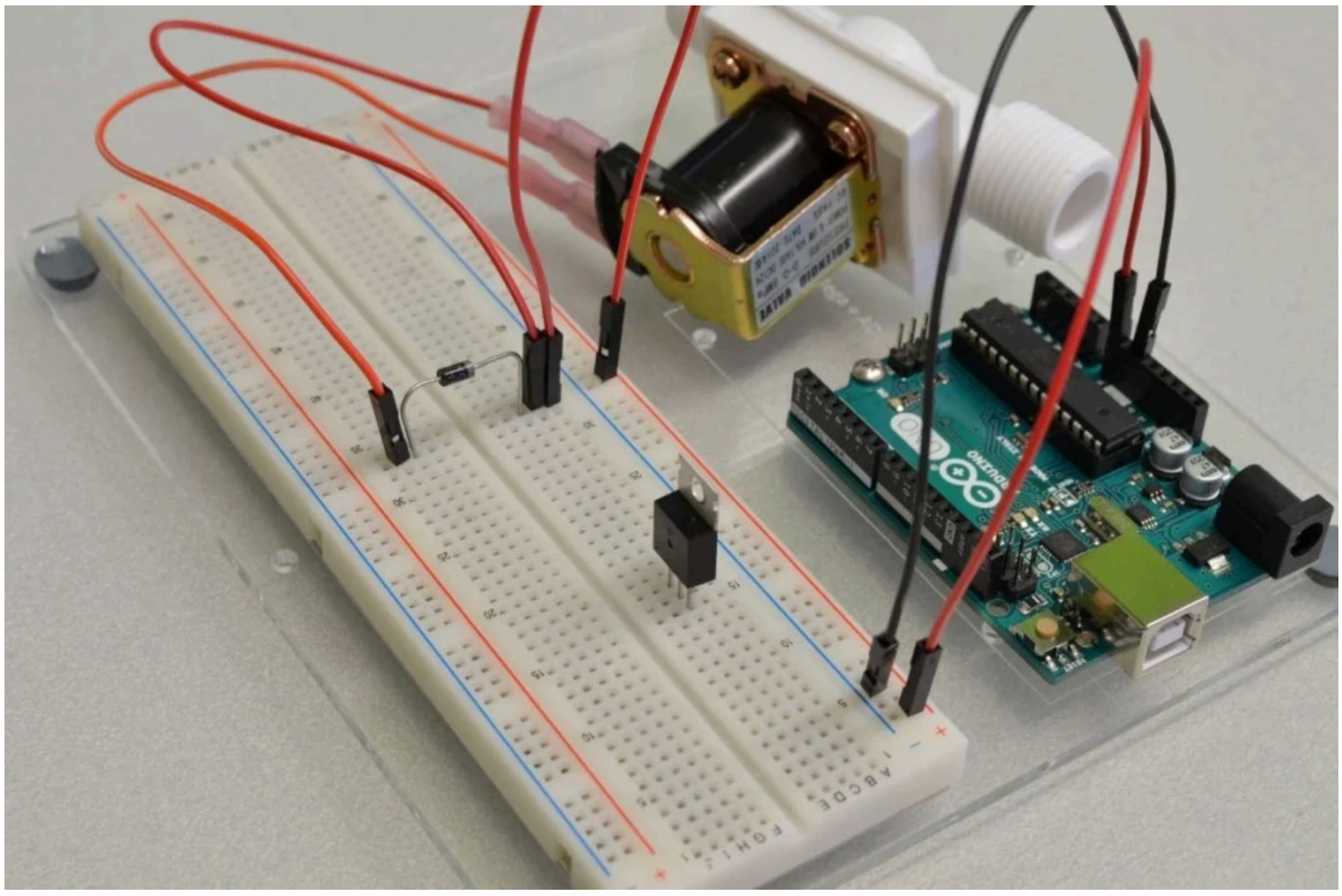
Step 5 - Power To The Solenoid



Now that we are sure the diode is facing the correct direction the solenoid can be attached to the 9V Power on the breadboard. The solenoid gets constant power because we will using low side switching to turn on and off this solenoid. Low side switching means we will be interrupting the circuit between the negative side of the solenoid and the ground rather than between the power and the solenoid. This seems a little counter intuitive, but we do this because switching the high side is a lot more difficult with a transistor when the voltage being switched is higher than the Arduino's 5V logic. Don't worry, in later tutorials we will go into high side switching!

33.3%

Step 6 - The Transistor

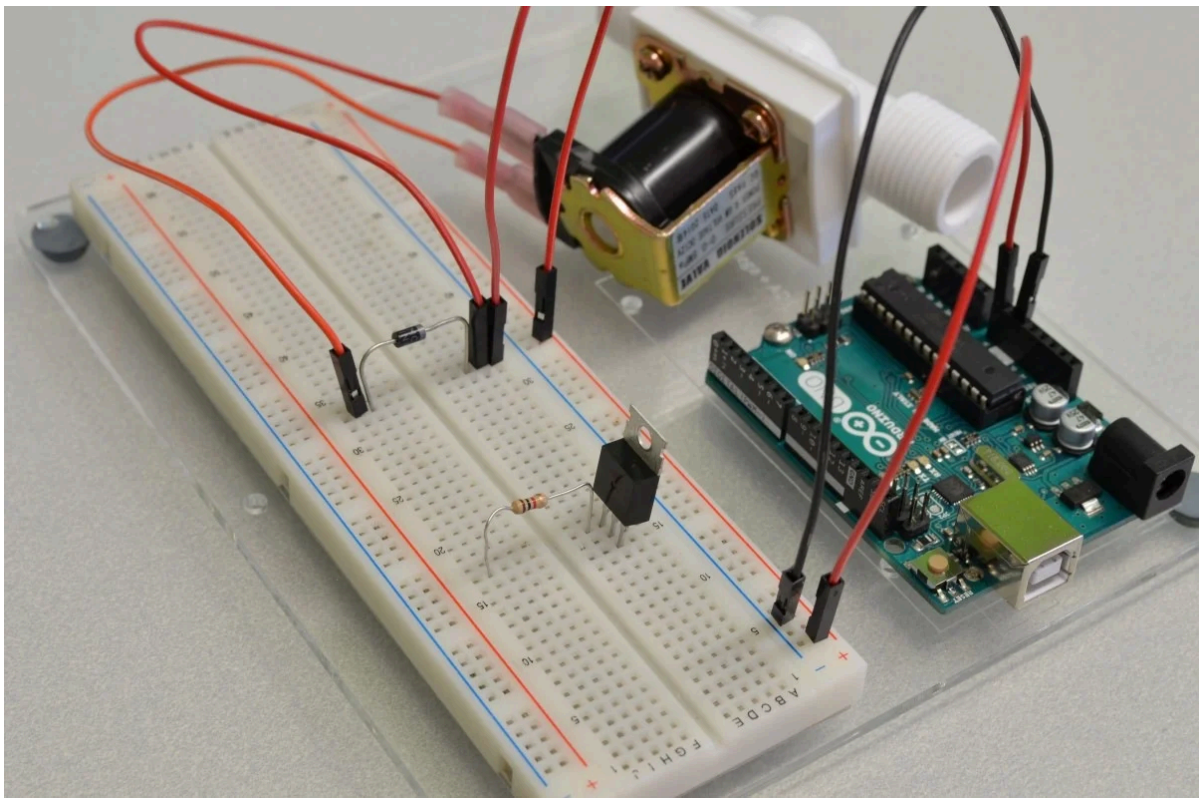


The current draw of this solenoid is higher than a standard transistor can handle so we will be using a TIP120 Darlington Transistor. A Darlington transistor is actually a pair of transistors that act as a single transistor with a high current gain. The pin output is still the same as a standard transistor so (for now) just think of this as a transistor with a higher current rating.

We will start by placing this transistor in the breadboard.

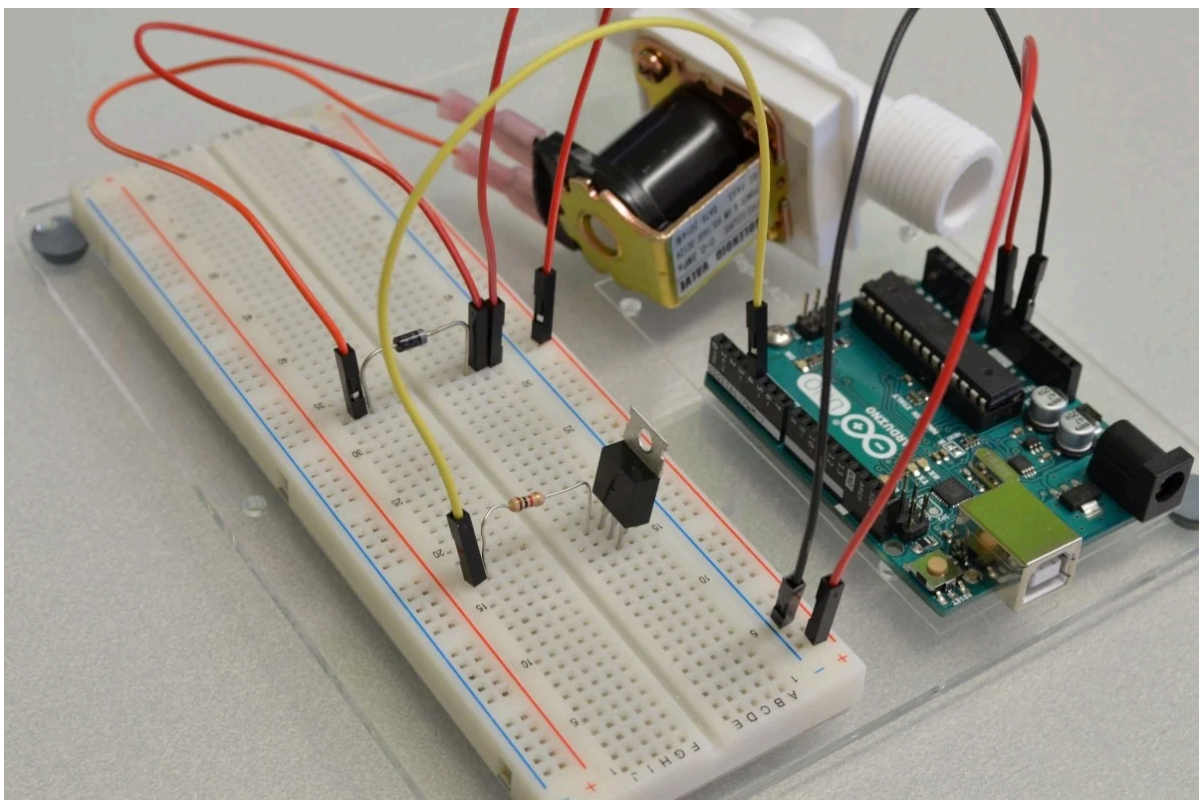
40%

Step 7 - Base Resistor



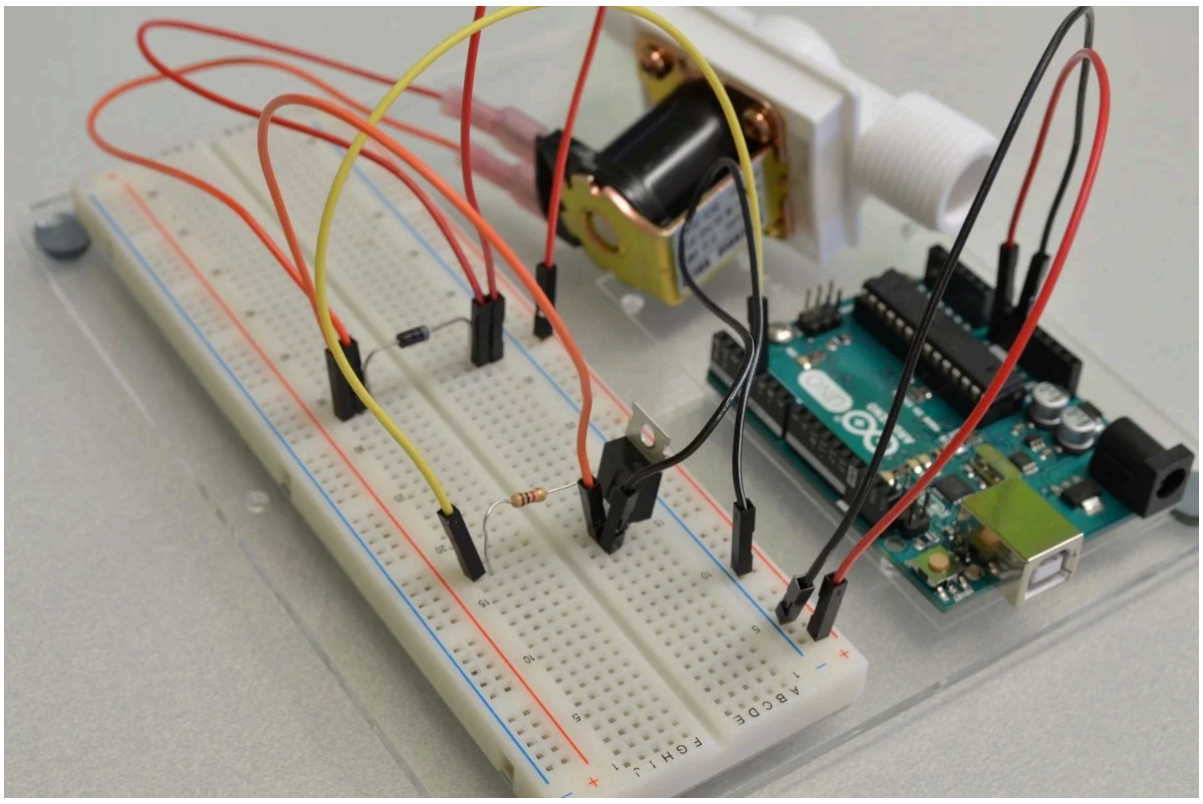
A base resistor is exactly what it sounds like – it is a resistor placed on the base pin of the transistor. This resistor limits the current going to the base (control line) of the transistor; no resistor would result in no current limit, and could result in a transistor blowing up! We will be using a 1K ohm resistor in this case; it can be placed from the base of the transistor as shown.

Step 8 - Connecting To The Arduino



Now that the current limiting resistor is in place we can go ahead and connect this up to one of the Arduino digital pins. Take a wire and run it from Arduino pin 4 to the current limiting resistor we just placed in the last step.

Step 9 - Connect The Solenoid



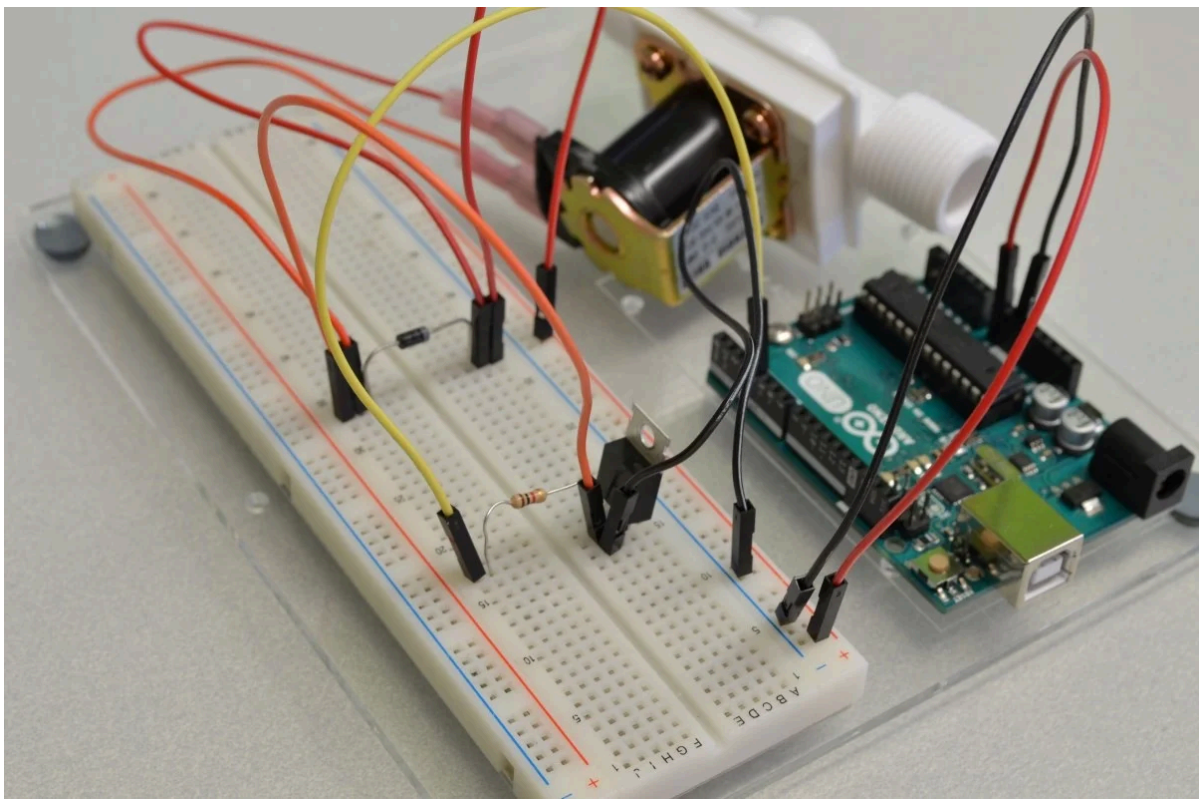
Next we are going to connect the solenoid's negative terminal to the collector on the transistor. The collector is one side of the "switch" in a transistor, this is connected to the emitter (other side of the "switch") when the base pin is has a voltage applied. An easy way to remember what goes where on a transistor is:

"The Collector collects whatever the Emitter will emit when the Base commands it to"

So in this case we are going to "collect" the negative from the solenoid and "emit" it to the ground of the circuit. So let's run a wire from the solenoid negative to the middle pin (collector) of the transistor.

60%

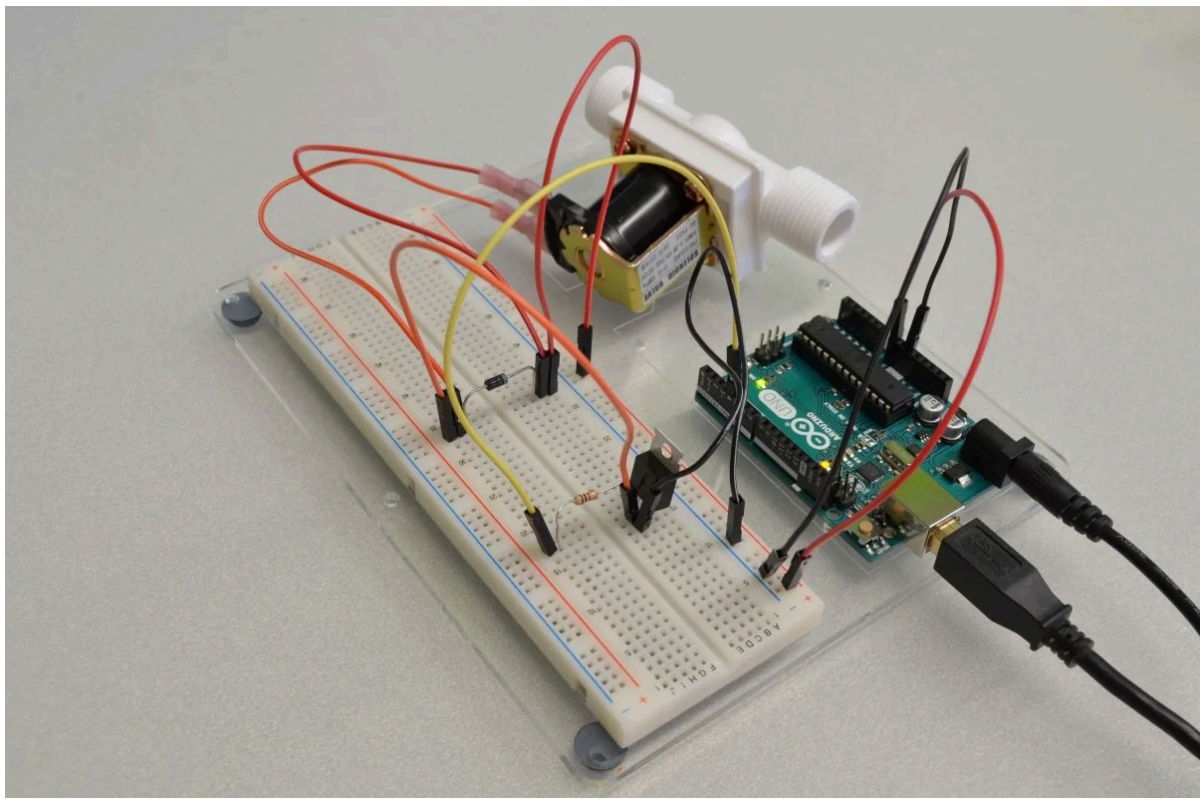
Step 10 - Connecting To Ground



Now we will connect the transistor's emitter to the ground rail on the breadboard. The circuit is complete!

66.7%

Step 11 - Double Check And Plug It In!



Before we give the Arduino power it is always a good idea to go over all of the connections to make sure there are no wires in the wrong spot – sometimes that can make for a very expensive mistake!

One way to avoid this problem is good wire color discipline. In other words, decide on a purpose for each color of wire and stick to them! In this example all 9V power are red wires, all grounds are black wires, orange is the negative output of the solenoid, and yellow is the signal wire. This way, if you ever see a red wire going to a black wire you will know right away that something isn't quite right!

73.3%

Step 12 - Starting The Code

```
1  void setup() {
2  // put your setup code here, to run once:
3  }
4  void loop() {
5  // put your main code here, to run repeatedly:
6  }
```

Now that we have finished with the hookup we need to start writing some code. We will be using the Arduino IDE, this is available from https://www.arduino.cc/en/Main/Software

We will start with the "BareMinimum" sketch found by clicking "File" and selecting Examples / Basic / BareMinimum. This sketch is a great starting point as it includes the Setup and Loop functions – we will write the rest!

80%

Step 13 - Understanding How To Control The Solenoid

Since the transistor is doing all of the heavy lifting in this circuit we do not need to do much in terms of coding. When Arduino pin 4 is set to HIGH this will connect the transistors collector to the transistor's emitter, which will activate the solenoid.

Let's go write the code!

Step 6 - Writing The Code

We are starting with the BareMinimum Sketch found in the IDE, it should look something like this:

```
void setup() {
// put your setup code here, to run once:
}
void loop() {
// put your main code here, to run repeatedly:
}
```

So first we will need a variable for the Arduino pin:

```
int solenoidPin = 4; //This is the output pin on the Arduino we are using

void setup() {
// put your setup code here, to run once:
}

void loop() {
// put your main code here, to run repeatedly:
}
```

Next we need to set the Arduino pin to act as an output:

```
int solenoidPin = 4; //This is the output pin on the Arduino we are using

void setup() {
// put your setup code here, to run once:
pinMode(solenoidPin, OUTPUT); //Sets the pin as an output
}

void loop() {
// put your main code here, to run repeatedly:
}
```

Ok, now all of the pins are set to outputs. Next we can write some code in the loop to switch a relay on and off:

```arduino
int solenoidPin = 4; //This is the output pin on the Arduino we are using

void setup() {
// put your setup code here, to run once:
pinMode(solenoidPin, OUTPUT); //Sets the pin as an output
}

void loop() {
// put your main code here, to run repeatedly:
digitalWrite(solenoidPin, HIGH); //Switch Solenoid ON
delay(1000); //Wait 1 Second
digitalWrite(solenoidPin, LOW); //Switch Solenoid OFF
delay(1000); //Wait 1 Second
}
```

So if we want the solenoid to allow water to flow, set the pin high. When you want the water to stop flowing, set the pin low. In this case it will turn the water on for 1 second and then off for 1 second, looping forever (or at least until it is unplugged!) This solenoid valve could easily be used with the flow meter featured in our last tutorial to create a system that only allows a certain volume of water to flow before shutting off.

93.3%

Step 15 – Upload The Code And Test



Now that all of the code has been written it can be uploaded to your Arduino! Click “Upload” button in the top left corner of the Arduino IDE and it should upload without any issues. After a few seconds the solenoid will start opening and closing.

100%

187 THOUGHTS ON “CONTROLLING A SOLENOID VALVE WITH ARDUINO”

MARK

Reply

September 7, 2015

umm, i wanna ask… how about controlling 3 solenoid valve? can it still be the same way to control 3 solenoid valve? is there any other step to control 3 solenoid ?
can you help me solve this problem



CHRIS @ BCR

Reply

September 9, 2015

Hi Mark,
Do you have a link to the 3 solenoid valve? Without seeing it, it would be a guess… likely it would need the same circuit for each coil.

JAKE

Reply

December 4, 2016

You could use a ULN200A Darlington transistor array to that project.
Hope this helped

MARK

Reply

September 11, 2015

so, i have to make the other two solenoid into paralel?
every website/ link i have been seen .., they only control 1 solenoid valve

---



CHRIS @ BCR

Reply

September 11, 2015

Possibly – but if you have a picture we could answer that with a little more certainty 😀

MATT

Reply

January 4, 2016

Could I separately power the breadboard (say 12v from a separate supply) and still do low side switching? I'm completely new to the world of electronics and don't want to destroy my Arduino from a rookie mistake. Essentially I wish to power several solenoid valves via the same breadboard and it would likely require more current than the Arduino can provide to the breadboard, hence the separate power supply.



CHRIS @ BCR

Reply

January 4, 2016

Hi Matt,
The pin we are using to bring power to the breadboard is labelled "VIN" and stands for "Voltage Input" on the Arduino – this is a direct pass through from the DC Barrel Jack on the Arduino. We used a 9V power supply in the tutorial but you can use a 12V power supply with that same DC barrel jack. Adding additional solenoid valves would not result in ruining the Arduino as you are directly powering the solenoid valves with the wall power supply already. Let us know if you have any additional questions!

MATT

Reply

January 4, 2016

Thank you so much for the quick reply!
The project that I have been thrown into would involves a logic system that would allow for me to have three solenoid valves open at the same time on timed intervals and remain open for a length of time. My concern is finding a DC barrel jack that would have the amperage needed to hold three solenoids open, especially since the system is gravity fed with negligible

pressure resulting in (oddly, or so far as I can find) solenoids that require higher amperage to operate. For example I'm looking at 3 valves requiring 3 amps each to operate.
Assuming I could find a barrel jack power source with a high enough amperage to operate the valves (9 amps or greater), you're saying I could just use the VIN terminal instead of powering the board separately? And the wiring system used in this tutorial could be replicated three times (once for each solenoid) to operate the system? Excepting the coding of course which I have experience with.

CHRIS @ BCR Reply

January 4, 2016

Hi Matt,
At 3A per solenoid you may want to take another approach all together. The solderless breadboard will not handle the total power of 9A, nor will the trace on Arduino. 3A per channel will exceed the capabilities of the TIP-120 as well. While there are better technical solutions to this problem – if you would like to build it on a breadboard I would recommend getting it all up and running with relays in place of the solenoids. The relays would act as on/off switches for the high powered solenoids. If you want to get in touch with us regarding the details of your project please feel free to email us at info@bc-robotics.com

SHAWN Reply

April 19, 2017

So an external power supply of 9v was used on breadboard all along? All along from my understanding i thought Vin from arduino would provide the 9v for solenoid valve. Please clarify this.

CHRIS @ BCR Reply

April 20, 2017

Ultimately, whether you supply the 9V through the breadboard or the DC barrel jack on the Arduino, the end result would be the same. In the last photo you can see we are using the 9V power supply in the DC barrel jack on the Arduino (and this is done simply out of convenience). If you were to feed 9V to the breadboard the Arduino would just be powered by the VIN pin rather than sending power to the breadboard by way of the VIN pin. The important thing to understand is that the VIN pin and the DC barrel jack on the Arduino are tied together directly; this VIN bus is also tied to the built in 5V regulator on the Arduino that powers the Arduino. As long as the VIN bus has 9V on it the Arduino should get its power and everything should work properly.

BART GREFTE Reply

January 20, 2016

"The inlet water pressure actually holds the valve closed so if you do not have any inlet water pressure (3psi minimum) the valve will never close!"
Can you recommend a valve that does not have that limitation?

CHRIS @ BCR Reply

January 26, 2016

Hello Bart,
We do not have a product without that limitation unfortunately.

MIKE Reply

November 17, 2016

Look into latching valves. They use a small current to open and a reverse polarity to close. You could open and close them one at a time.

CHRISTIAN Reply

January 19, 2018

Fast Same Day Shipping

2000+ In Stock Items

Ships From Canada!

In-Depth Tutorials

I will be using 2 solenoids for our project. Do we need to have a relay or can we avoid using a relay? And how do we go about that? I am having challenges with connecting it to the breadboard.

What is BC Robotics?

BC Robotics Inc. is a Canadian owned electronics company based in Nanaimo, British Columbia. We manufacture 80+ different electronic accessories and stock 2000+ unique and interesting electronics from popular brands including Arduino, Raspberry Pi, BBC micro:bit, Adafruit, SparkFun, Makey Makey and more! Fast Shipping – Orders placed before 3PM Pacific Time ship out same day!

CHRIS @ BCR

January 26, 2016

Reply

A relay would not be necessary unless the power requirements of the solenoid exceed the amount of power the Darlington transistor is able to switch.

secured by

LIONEL EISENBERG

January 26, 2016

Reply

Hi Chris I have been trying to make the board work for some time now but for some reason im not getting any voltage, ie if i plug a voltmeter where the solenoid is supposed to be I get 0V could you help?

Company Information

About Us

Frequently Asked Questions

How To Order

Shipping Guidelines

Track Your Order

Become a Distributor

Terms Of Service

Privacy Policy

Contact Us

CHRIS @ BCR

February 9, 2016

Reply

Hi Lionel,

Please post (or send) us a photo of what you have so far!

JOSH

February 9, 2016

Reply

Hello!! I have an Arduino Uno and a 5v solenoid. Is there any way I can power it without frying the unit? I am completely new to this kind of stuff and looking for any guidance on how to approach this. Thanks so much in advance!

Subscribe To Our Newsletter

Get the latest news on Promotions, Sales, Coupon Codes, New Product Updates, and more delivered directly to your inbox.

Email Address *

Subscribe

We don't spam! Read our privacy policy for more info.

CHRIS @ BCR

February 9, 2016

Reply

Hi Josh,

Rather than a 9 or 12V power supply, switch it out for a 5V supply to power the entire thing. The wiring will stay almost identical – just swap the red wire powering the Arduino from the "VIN" pin to the "5V" pin.

JOSH

February 9, 2016

Reply

Alright awesome, thanks for the quick reply. Would I need to change the type of transistor or diode? Thanks again!

Copyright © 2026 BC Robotics Inc.
All rights reserved.
Nanaimo BC, Canada

February 9, 2016

Also, last question… I promise! If I have this 5v solenoid, https://www.adafruit.com/products/2776 … How would I connect the 2 pin to the breadboard? Alligator clips? I'm completely sorry but I have no idea how to work this!!!

CHRIS @ BCR Reply

February 10, 2016

Hi Josh,
You would not need to change any of the components as the circuit is switching the ground. As for the solenoid – I would suggest using a chunk of wire just like they have done in the photo. If you do not need that connector on the end you could always chop it off as well.

JOSH Reply

February 11, 2016

Alright awesome, thanks so much again with help. My solenoid arrives Tuesday, so I will let you know how the setup goes. thanks!!

NAGESH Reply

February 21, 2016

I am using 24v solenoid water valve… but i am unable to connect to arduino uno as shown in this page wiring diamgram. Please advice whether i should used relay to connect 24v solenoid to arduino?

AJAY Reply

March 7, 2016

Hi Chris,
I am using three 12v DC push pull magnet 7W for my project. I have hooked my arduino to big easy driver to control the stepper motor while on the other pins I'd like to control these 3 cylindrical magnets to go on&off for a period of time.
My question is should I use relay modules rated 5v –12v DC to control these magnets or should I go for transistor and diodes like you illustrated?
I'd be obliged if you can help me with my question.

SERGIO Reply

March 25, 2016

Hi, It's possible to control 100 solenoids? I'm a programmer and not electronic engineer :/
What changes would I have to make so 1 arduino could control 100 solenoids?
Thanks

SABRI Reply

May 2, 2016

Hi, is this solenoid only match with TIP120 Darlington or can use others? Recently I used TIP42C, my program is not giving an output.

CHRIS @ BCR Reply

May 3, 2016

May 5, 2016

Hello,
A TIP42C is a PNP Darlington, in this circuit you would need an NPN Darlington (So a TIP41C should work)

VWHEATLE Reply

June 6, 2018

Could a TIP110 transistor also work? What is the current that the solenoid will draw?

DAVID Reply

May 27, 2016

Hey guys,
I have a similar issue to some of the above comments, have read through them but am still unsure due to my inexperience.
I'm trying to operate 5 different 12V water solenoid valves through a Arduino Uno. How would I go about powering this kind of project?
Obviously once the powering and layout is done I'm assuming the code is just like the one showed, so i'll keep my question simple 🙄
Thanks in advance!

CHRIS @ BCR Reply

May 30, 2016

Yes, you would power it the same as above – the diagram above will handle as many solenoids as you have IO pins available.

ADI Reply

June 11, 2016

Thanks for your awesome tutorial. I'm totally new to arduino. 😀
Quick question here, can I use Vin pin to power a solenoid as in your tutorial while at the same time using the 5V pin for another purpose?
Big thanks!

CHRIS @ BCR Reply

June 23, 2016

You can still use the 5V pin because it is pulling power off of the 5V regulator on the Arduino *but* I wouldn't be powering anything too power hungry with it. You can always use something like an LM7805 to create another 5V source on the breadboard.

MAHESH Reply

June 19, 2016

Great tutorial. . Finally everything is working .. I added few switches to control flow etc …
I used uno r3 .. plan to go for yun as it can connect to AWS IOT ..
However found that yun does not have DC input. So how will circuit look like with yun .. will I need a relay .. or can I still use transistor approach ..
hard to digest if I can not use this thing ..
Thanks

CHRIS @ BCR Reply

June 23, 2016

Hello,
With the Yun you would need to add in a 5V regulator on the breadboard to feed the Arduino. Alternatively you can power the Yun from a separate power supply as long as you tie the grounds together.
Hopefully that helps!

SANJEEV KUMAR Reply

November 17, 2016

on Yun you can power it from VIN pin on the analog side.

ARNAV RAJ Reply

July 2, 2016

Can I power the solenoid by using only the USB connection in arduino.

MIKE87 Reply

July 16, 2016

no it wont work

RCNAME Reply

November 3, 2018

How I power the selenoid then?

SAPWOOD Reply

December 23, 2020

for the next person that reads this, you use a 5v or 12v dedicated power supply for the arduino and use the VIN pin location to supply that voltage to your circuit. if you use a 12v power supply the arduino steps it down for itself and passes the 12v around to the VIN pin…
or so i think that's how it works

CAMERON JAMES Reply

July 5, 2016

To whom it may concern,
I am working on an Arduino project that incorporates both a solenoid valve and a liquid flow meter, which I want to have activate at the same time. I viewed the tutorials for each of these on the BC Robotics website, and I noticed it was mentioned that these could be easily used together. Any help or guidance with how wiring would work to operate both of these at the same time would be much appreciated. The solenoid valve and liquid flow meter I will be using can be viewed in the links below:
https://www.adafruit.com/products/996
https://www.adafruit.com/products/833
Thanks

DIEGO GARCIA Reply

July 10, 2016

Thanks for the tutorial.
I am having a little trouble with it however.
I have done everything correctly including the code but the solenoid valve wont open.
I used a voltmeter to make sure I was getting current at the valve and I am, but just around 6 volts.
The battery is brand new so I don't now why the voltage is that low at the valve.
Any help would be appreciated.

DIEGO GARCIA Reply

July 10, 2016

Also when I disconnect the usb from the Arduino, I will only get a current for an instant and then nothing at all.

MIKE87 Reply

July 16, 2016

what is the battery rating?

LUKE HARRISON Reply

August 9, 2016

Hi, friends and I working on getting a solenoid valve to work for a school project, and we have done everything according to your tutorial and it doesn't work.
The valve only opens when we blow to create pressure, otherwise no amount of water in a tank (that is directly connected to the valve) is able to open it.
We tried connecting the power to 5V but it made no difference.
We would greatly appreciate your help,

CHRIS @ BCR Reply

August 11, 2016

Hi Luke!
The issue is likely not with anything you have wired or programmed – the valve we carry needs minimum of 3PSI of pressure to open.

LUKE HARRISON Reply

August 12, 2016

Ah, ok. Thanks!

ADAM GOODMAN Reply

January 12, 2017

You would need 7ft of water level above the valve to create 3psi of pressure at the valve.

ASH Reply

August 12, 2016

Hi there, thank you so much for this tutorial, it is very informative.
I was wondering, for my project, I need to connect a 24V solenoid valve to Arduino, is that possible? what modifications would I need to make to the circuit or components (diode, transistor,resistor)?

CHRIS @ BCR Reply

August 17, 2016

I would first check and see if 12V will trip the solenoid reliably… if so, just do that!
Otherwise the biggest thing you need to worry about is powering the Arduino. The easiest way would be to power the Arduino from a separate power supply. Everything else will tolerate the 24V power.

ROBERT KIPNGENO Reply

August 24, 2016

Hi what of air tight solenoid valve? Does it use the same circuit ,do you stock them and how much are u selling ?

MANDI @ BCR

Reply

August 27, 2016

Yes, it does use the same circuit. We carry this valve which might work with your application.

GREG

Reply

August 31, 2016

Greetings,
How close does the solenoid need to be to the board. I am putting this in a high tunnel and will have to protect it from the elements.
Thanks,
Newbee

CHRIS @ BCR

Reply

September 6, 2016

It doesn't need to be that close – but ultimately it will have more to do with the gauge of wire than anything else.

YOSSI

Reply

September 5, 2016

Thanks for the tutorial!
Is is possible to to turn on and off the solenoid only with a relay board?
If yes i would like to know if needed yo add diodes or resistors?
If not i would like to know why?
Thanks for your help

CHRIS @ BCR

Reply

September 6, 2016

In theory you could use a relay – the only reason to do this would be if the Solenoid is powered by a different power source (Either AC or higher voltage, etc). As for the additional components: whether or not they are needed and the type of component needed will always depend on how you are hooking it up and how you are powering it – there is no generic answer.

VICTORIA

Reply

September 7, 2016

Hello. I used a solenoid directional air control valve instead of the the same solenoid valve that you used and did the same connection. I could see the LED light inside turn on/ off but the valve itself was not open/ close properly. Do you know why? The solenoid I used is 12V rated. Thanks a lot.

CHAVI

Reply

September 8, 2016

Hi,
I have a solenoid that's not close automatically, then I need to invert polarity to close, I tried with an H-bridge but it's not working. If I follow the tutorial and I change the wires the solenoid opens and closes correctly. Why it's not working with an h-bridge? and what can I do?
Thanks!!

CHRIS @ BCR

Reply

September 12, 2016

The H-bridge is probably the best way to approach it – We will have a tutorial for using an H-bridge coming up soon. In the mean time I would suggest throwing a motor in place of the servo and see if it works in both directions.

HAFIZ

Reply

September 11, 2016

Hi Chris,

Thanks for the tutorial. Is there any way that I can control 2 solenoids at the same time?

CHRIS @ BCR

Reply

September 12, 2016

The easiest way to accomplish this would be to parallel the entire circuit, and just connect the base of the transistor to another one of the Arduino's digital I/O. Just modify the code to trigger the correct valve at the correct time and you should be good to go!

SIDRAH

Reply

January 18, 2018

I'm quite new to coding, so how would I modify the code to trigger the correct valve. Would be great if you could reply

BABAK

Reply

September 11, 2016

hi, would like to know what is the maximum length of wire i can hook up to this solenoid to work with arduino?

MIKE87

Reply

September 12, 2016

It depends? What is the input voltage tripping the solenoid, the linear resistance of the wire, the ambient temperature, current draw of the solenoid, trigger voltage of the solenoid, and so on.

ABHIRUP DAS

Reply

October 2, 2016

I think a mistake has been made.

In Step 9. The last paragraph.

You have written.:"So let's run a wire from the solenoid negative to the middle pin (emitter) of the transistor."

The term in brackets should be "collector" in the middle pin of the transistor.

Great tutorial tho for a beginner. Cleared lots of queries! 😀

CHRIS @ BCR

Reply

October 2, 2016

Thank you for pointing that out, it is a good catch! We have updated the tutorial to fix the flipped pin name.

JOSHUA

Reply

October 7, 2016

Thanks for the tutorial!

I feel like I am missing something, is there supposed to be a 9v power supply plugged into the dc barrel jack in the tutorial? I don't see that in any of the pictures, but otherwise where does the power come from? I am trying to power a 12 volt 7 watt solenoid valve so would need to trade the 9 volt power supply for a 12 volt power supply. Also, would this cause any problems? Based on your previous comment it seems like this should be fine.

CHRIS @ BCR

Reply

October 12, 2016

A 12VDC power supply will work as well

HERNAN

Reply

October 7, 2016

Hi! Excellent explanation! I am designing the same circuit but the solenoids I have are 1A. Since my strength is programming and not electronics, I have some doubts about the values of the components.
Is the 1N4001 doide enough? (I was thinking about replacing it with one that tolerates up to 3A). Would the TIP120 tolerate 1A? Do I need a bigger resistor?
I would be thankful if anyone could provide any help!

HERNAN

Reply

October 7, 2016

(Power supply: 12V 3A)

S. S. AKSHAY

Reply

October 12, 2016

Is water supply mandatory? Because the valve isn't closing at all. It's quite urgent, help!

EDWARD DILLER

Reply

December 24, 2016

Hi, what would I have to change if I wanted it to be powered by a 12V battery?
Thank you!

CHATTIP

Reply

January 17, 2017

Hi, I'm developing a CO2 level control. I got an MG-811 and want it to connect to a 12V solenoid valve to be opened and release CO2 gas from the tank when the level is below the set point, then closed when it reaches the limit.
Please suggest how I would go from this excellent demonstration. (I'm not in this field but need to use this for my research)
Thank you so much in advance.

GIOVANNI NIEDDU

Reply

March 14, 2019

I have the same your problem, have solve?

KHEITO

Reply

January 22, 2017

Can I use the cellphone charger (NOT the usb one) as an external power supply of the arduino? Thanks in advance!

NICOLO TANTI

Reply

February 28, 2017

Hello, I would like to control a double-acting pneumatic piston using a 24Vdc solenoid valve using relays on both ends. What modifications are

required to drive the circuitry due to these changes please? Thanks in advance

HUMERA Reply

March 8, 2017

Does the same code works for the solenoid 3valves DC-12V???I tried working with the same code but it's not showing any effect on the valves. Can u plz help out.

EVIE Reply

April 5, 2017

My solenoid will not loop or move on its own. When I push the shaft it snaps in, waits the programed delay period, and then pops out. It will not loop or pull on its own, but it is responding to the code. Any idea what is going on?

PRATEEK SHARMA Reply

April 7, 2017

what happen if we connect 12 volt supply to this circuit.

CHRIS @ BCR Reply

April 20, 2017

As long as 12V works with the solenoid, it will not cause any problems.

YIJIA SUN Reply

May 9, 2017

Hi. I am wondering what I need to add if I want to control 8 solenoid liquid valves or solenoid gas valves at the same time?
Can the solenoid valves on this page achieve that?
Do I need a larger solderless breadboard or just change another transistor?

RAVI Reply

May 19, 2017

Can i use IRF 540 or any other MOSFET instead of using TIP120, the latter is unavailabe in my locality right now.

MIKE87 Reply

May 31, 2017

You could use the IRF540. Just make sure that the gate / source / drain are hooked up right.

JULIA Reply

May 29, 2017

How do you attach 2 solenoids in parallel and have a button for each, allowing to turn the solenoids on when you press the buttons.

MATT Reply

June 4, 2017

If I where to use a RPi which has the 3v pin, what chips would I used?

MATT

Reply

June 4, 2017

5v*

MIKE87

Reply

July 8, 2017

You will need to move the 12v power supply to the breadboard and just connect the ground between the breadboard and the ground pin on the pi. The Pi would run on its normal power supply.
The digital pin part of the circuit will still work the same even though it is 3.3v instead of 5v

---

SASH

Reply

July 8, 2017

Hi,
I'm new to all this so apologies for the basic question. I want to replicate this for six solenoid, and have done this on a breadboard. When I test, only the one closest to the power supply works. I am currently running off the arduino power supply to test, is this because the current is too low or do I need to change the circuit? I will use An external 12v wall plug in the live version.
Thanks for the great tutorial!
Here's the circuit:
https://goo.gl/photos/UH2zt5zfNpRdeZfN6

MIKE87

Reply

July 8, 2017

use a multi meter and check to make sure there is power getting to each of the solenoids. Also check continuity for all the grounds.
If all that checks out test the signal is actually being sent from the arduino.

PETER

Reply

July 11, 2017

Thanks for the detailed tutorial!
I've been following the steps to the letter, however unfortunately it the valve doesn't seem to be doing anything. For reference this is my set up:
https://i.imgur.com/qtLK3RG.jpg?1
I am using this transistor [pdf]: http://www.onsemi.com/pub/Collateral/P2N2222A-D.PDF
This diode: https://www.sparkfun.com/products/8588
And a 330ohm resistor instead of 1k. I have a 330 or 10k available, but figured the 10k might be a bit a much.

JO

Reply

July 16, 2017

Thanks for the very good tutorial. I got it to work and i never did electronics before.
What would be the best alternative to chose from when i want to control the valve based upon a timeschedule. Is there some software i can download?
Thanks

GAGE

Reply

July 28, 2017

Can solenoids be controlled with a potentiometer to open proportionally?

GAGE

Reply

July 28, 2017

also, which solenoid valve would be best for propane/butane? I have been searching for one for a few days now but haven't found one small enough to be used in my project. It needs to be close to 3×3″ but can be smaller.

SUMEET Reply

August 11, 2017

how to control it with bluetooth. pls help.

JOHN COSSO Reply

October 18, 2017

Hey my arduino and valve are set up the exact same as yours but mine valve makes a tick feel instead of completely opening, do you know what the problem is?

MIKE87 Reply

October 21, 2017

I bet it is your power supply – make sure it is powerful enough to trip the solenoid by running the power and ground wires from the power supply to the solenoid. It should open as soon as you connect them.

BEARDOFZEUS Reply

March 29, 2021

I had a similar issues and was using 6 AA batteries to power the board and solenoid.
When I changed the circuit to switch to a dedicated 9V battery on the TIP120 and used the 6x AA's to power the board, the solenoid opened properly.
The voltage drop across the TIP120 was too large when using 6x AA's so didn't have enough voltage at the solenoid to actuate it. Hope this helps you and maybe others!

GOBOT2GO Reply

November 10, 2017

Is there a way to make this same circuit work with an Arduino Nano?

NELSON Reply

November 13, 2017

Hi can we use a p55nf06 power mosfet in the place of the TIP120 ?

CHRIS @ BCR Reply

November 14, 2017

You could use an N channel MOSFET – just be sure to check the data sheet for the part to make sure it is connected correctly

---

JOHN Reply

October 31, 2018

Using a MOSFET ie IRF830 transistor. How are the connections like? For TIP120 its the base controller emitter but IRF830 is gate drain source. What is the equivalent?
Thanks in advance

MATHEUS

Reply

November 21, 2017

Hello Chris! Thanks for the tutorial!
Got everything set but it isn't working. The valves aren't doing anything.
Tested Arduino and it is working okay but the solenoid isn't doing a thing. Checked the wires again and again.
Any idea on how to proceed?
Thanks in advance!

NIRMALARAJAH

Reply

December 6, 2017

Why want to use this(1 x TIP120 Darlington Transistor)
particular transistor for this circuit?

NIRMALARAJAH

Reply

December 6, 2017

How did u give 9V to breadboard through Arduino??

EDUARDO GARCETE

Reply

December 21, 2017

Hello, thanks for this very useful tutorial!, but Im having a trouble.
Im using six 5V small solenoids, what changes should I consider to do?
Actually I did use this same circuit, with the same components and it works, but the solenoids seem to overheat quickly, or it is normal?. I've tried to use 5V pin instead of VIN but it doesnt power up enough all the solenoids.
Sorry for the noob question, Im new in electronics and sorry for the english in advance.
Any help will be appreciated!

ROHAN

Reply

February 13, 2018

Hello,
Can the solenoid push headphone jack into laptop/TV
Thanks.

MIKE87

Reply

February 14, 2018

only one way to find out!

DAN

Reply

February 21, 2018

Is it possible to use PWM to control the water rate using this valve?

RAJAGOPALAN

Reply

March 11, 2018

Can I use 7805CT in place of TIP120?

CHRIS @ BCR

Reply

March 12, 2018

You cannot – a 7805 is a voltage regulator while a TIP120 is a Darlington Transistor

---

ASHWAN VISRAM

Reply

March 12, 2018

Hello.
I followed each step and used exactly each of the components listed. The solenoid valve is not responding as all.
Can you please help?

RAJAGOPALAN

Reply

March 13, 2018

same here

MIKE87

Reply

March 19, 2018

does the solenoid work when you connect it up to the power supply?

RAJAGOPALAN

Reply

March 13, 2018

I am using solenoid valve A type DC 12 230mA. KNF18C06-10136
Diode 1N4007
Transistor TIP122/102
9V battery.. its still not working.
solenoid is perfect as it working when connecting directly.
any help you can do? thanks in advace

CHRIS @ BCR

Reply

March 19, 2018

The tutorial is designed to explain how to use the parts we sell – since those are not the same parts, you will need to check the specs on each individual part and compare them to the ones we used. If they are different in any way you will have to adapt your build accordingly.

JOSEPH

Reply

March 19, 2018

Do you need both the power (from the black outlet) and the USB power to ensure that it works or is USB power sufficient?

MIKE87

Reply

March 19, 2018

USB is only 5V and the solenoid is 12 – def. need the external power supply

---

VISHAL

Reply

April 7, 2018

Hi I am using the same parts as mentioned in the tutorial, but when i on the system, i dont see any water flow. I double checked the solenoid by connecting directly to 9V power and that is working. Can you please help ?

WILLIAM @ BC ROBOTICS Reply

April 10, 2018

Send us an email at support@bc-robotics.com and reference the post along with your invoice number and we can look into it. If you have not purchased with us you should get in touch with whomever you ordered from.

DANE Reply

April 8, 2018

What it the pink material in the solenoid? like a rubber? thanks

WILLIAM @ BC ROBOTICS Reply

April 10, 2018

Hi Dane – are you referring to the connectors? They are an insulated spade connector.

RAVI Reply

April 9, 2018

I want to use arduino pro mini instead of arduino uno. What changes should i do?

Provided that arduino mini pro is 3.3 V – 12 V in range

WILLIAM @ BC ROBOTICS Reply

April 10, 2018

It would just be a matter of adapting the pro mini in place – the 3.3v logic level will not make a difference.

PENGO Reply

June 24, 2018

So, if i want to control 9 valves at once, would this schematic do the trick? https://imgur.com/a/XYQdp2i

Thx !

TREX Reply

June 24, 2018

Great tutorial.

PENGO Reply

June 24, 2018

Thx for the guide. My project involves 9 valves so i'm curious, would this schematic do the trick and control them all at once? I'm not that gifted when it comes to hardware…

https://imgur.com/a/XYQdp2i

CHRIS @ BCR Reply

June 27, 2018

You are very close – just tie the ground rails on the breadboard together (looks like the bottom ground rail isn't tied to the rest of the

circuit right now) – once that is fixed it should work!

PENGO Reply

June 28, 2018

I see where i got wrong indeed, i started the schematic on the bottom ground rail but in the end moved it to the top to make it more clear and i forgot to connect the ground rail indeed. One more question if i can; i see in the specs that the transistor can take a maximum of 60V. So it can be used to power a valve working on 24V DC i suppose? If i'm correct i'm also assuming that the arduino can't give that amount of juice so i'm gonne need to plugin extra power supply and if so, where should i put it? Many thanks !

BRANDON THOMPSON Reply

June 25, 2018

This was very helpful. How would i control the valve with a muscle sensor?

ISAIAH SIRKIN Reply

July 12, 2018

Okay, I've added the code, I've connected all the wires, but the Solenoid won't start? Is it a problem with the power supply or do I just use a different valve

BEARDOFZEUS Reply

March 29, 2021

I had a similar issues and was using 6 AA batteries to power the board and solenoid.
When I changed the circuit to switch to a dedicated 9V battery on the TIP120 and used the 6x AA's to power the board, the solenoid opened properly.
The voltage drop across the TIP120 was too large when using 6x AA's so didn't have enough voltage at the solenoid to actuate it. Hope this helps you and maybe others!

SAURABH KESARI Reply

October 3, 2018

When we are directly connecting this transistor cicuit on arduino without using breadboard arduino is getting reset after one toggle can u tell me the possible cause as well as solution also.

CHOU TZUYU Reply

November 22, 2018

HIya,
So I want to make a spot welder with 12V Car battery. I would be using 12V, 500 A solenoid to pass the current. My solenoid is different than what's been used for the tutorial. Here's the link to the solenoid. I want help with connecting this solenoid with my schematic that I have posted below.
Solenoid link: https://www.amazon.co.uk/Current-Solenoid-Contactor-Batteries-Isolator/dp/B076HFSFCP/ref=as_li_ss_tl?ie=UTF8&qid=1511055833&sr=8-5&keywords=12v+starter+solenoid&linkCode=sl1&tag=dinkylicom-21&linkId=79a514c8f8c7cfb3fad68123a88d168a
Schematic link: https://www.youtube.com/watch?v=nmVqUkNjubk
Provided that everything else works, I want to replace the transistor with N-Type MOSFET as my solenoid will draw higher current. Would that work if I change transistor with MOSFET and kept everything else same?
Any help is highly appreciated. Thanks

CHRIS @ BCR

Reply

November 22, 2018

We don't provide help with custom projects, only questions directly relating to the tutorial.

SOUMEN NAAYAK

Reply

November 23, 2018

Hi,
I am also working with a similar kind of project, In which I am using a linear movement push type solenoid which is pushing 5 kg wait for some time and then it moved back to original position after some time. I want to make this project battery operating for at list one month but I am not able to understand how big battery I have to choose and solenoid should be controlled by Arduino. question is how I will get a 5-volt operating push type solenoid and it should be able to push 3-5 kg weight? and operating current should be 150mA max. hardly I am going to use this solenoid for a whole day and each time solenoid working for 10 sec.

DAHLIA BINTI MOHD FAISOL

Reply

November 26, 2018

hi, i am not familiar with the coding for arduino. how is the coding to make the valve is on and off using button

---

ANDY

Reply

November 28, 2018

Hey, Love the article, but how would you do this four four pumps? can you do a tutorial? 🙄

BRI

Reply

January 21, 2019

Hi,
Can the arduino uno be used to control 2 seperate solenoids?

DEEPAN

Reply

January 29, 2019

THANK YOU, THIS TUTORIAL IS VERY USEFUL. BUT, PLEASE EXPLAIN THE POWER SUPPLY HOW YOU HAD SUPPLIED 9V TO THE SOLENOID.WHEN, THE ARDUINO PIN OUTPUT IS LIMITED TO 5V. THANKS IN ADVANCE.

DEEPAN

Reply

January 29, 2019

thank you any way i got the answer in your comment.

RLURKINS

Reply

January 30, 2019

how many amps does this run off of? I will be using a solar panel and a charge controller how do i hook this into it

---

PAMELA COOPER

Reply

February 2, 2019

Hi, If it'm controlling an air pump how do you recommend I set up the circuit?

LEOSENKO

Reply

March 3, 2019

Is there any solenoid valve for 3/16″ aquarium tubing? All of these 1/2″ NPS tubes are too huge to control flow to potted plants, not mentioning the complications with all the connectors…

CHRI LU

Reply

March 11, 2019

Hi guys,
I'm new to all this DIY programming stuff and want to build an automated system for supplying a definied amount of water to a storage tank.
Since you propose the following: "This solenoid valve could easily be used with the flow meter featured in our last tutorial to create a system that only allows a certain volume of water to flow before shutting off."
Can you tell me how to combine the sensor and the valve or give me some keywords what to look after?
Kind regards
Chris

MARY DILLON

Reply

March 12, 2019

Hello, I'm trying to work out the math with this circuit and I seem to be getting a larger than expected Emitter current. Has someone worked out the math to prove this circuit works? If so can you at least tell me the Emitter current you got. I believe we're using all the same components, the solenoid load is 320mA. Thank you!

GIOVANNI NIEDDU

Reply

March 14, 2019

I want to handle CO2, instead of gas, can you help me? I need the same components, but instead of your solenoid valve, the only difference is to use a valve for gas? Can you suggest me an appropiate valve?

NIKÉ

Reply

March 28, 2019

When I manually push the solenoid in, the pin shoots back out after a second, but the pin won't pull in automatically. What can be the problem? I've checked my connection over and over again and I can't find a mistake in that

---

ANDREA

Reply

April 3, 2019

I will try and use this tutrial, everything will be the same, except, instead of automatically turning off, like in the code, how can i modify the code, so for example when you press letter R on keyboard solenoid valve will turn on and letter T will turn the valve off?
Thanks in advance!

SAIF ALSULIMMANI

Reply

April 19, 2019

I am working on a project where I have to code a solenoid to push popcorn kernels down through a pipe.
My device will measure a lego coin and based off of that, the number of pushes needed is determined by that.
Does anyone know how to code that?
like some kind of while loop
while (k>o) // k is the coin value
{
k = number of pushes
}

for example if k is 6, I want the solenoid to go forward “push” 6 times!
THANK YOU

HARDOMOR Reply

July 29, 2019

Trying to control a 12V solenoid valve but although the LED in the Arduino board lights up with the correct timing, the valve only manages a barely audible “click, click” sound without it actually opening. Could it be that the 6 AA batteries connected in series to power the arduino are not providing enough current despite giving 8-9V of voltage? And if so, is that a problem of the TIP120 transistor, maybe it is a defective one? (the shop had only one left and no analogues when I ordered, so I had to make do with the single transistor, altough it not working was a worry)

HARDOMOR Reply

July 29, 2019

This is the valve being used, for air actually, it needs no minimum back pressure and works for air and water: 2W-025-08
file.seekpart.com/keywordpdf/2011/3/25/20113251736480.pdf

BEARDOFZEUS Reply

March 29, 2021

I had a similar issues and was using 6 AA batteries to power the board and solenoid.
When I changed the circuit to switch to a dedicated 9V battery on the TIP120 and used the 6x AA’s to power the board, the solenoid opened properly.
The voltage drop across the TIP120 was too large when using 6x AA’s so didn’t have enough voltage at the solenoid to actuate it. Hope this helps you and maybe others!

MARIANA Reply

November 27, 2021

Could you share the schematics you used to power the board and the solenoid with different power sources?

SURYA Reply

September 30, 2019

“I established Stepdown transformer to design own power supply to led lights design
I used Stepdown 12 volts tarnsformer and lm317 100 l variable resistor I connected the lights from transformer and other side I connected Arduino Uno with 5 volts charger but the program is executed in l board but other transformer side light was not blinking my doubt is both grounds are commonly not connected is that problem I feared about short circuit please give any answer how to turn on power supply thank you.”

JEENU Reply

November 12, 2019

Can I use a IRF510 transistor instead of the TIP120 transistor?

EDY Reply

December 10, 2019

hey how can u connect a PIR sensor with this? I’m making a project and it would really help if I knew

JASON Reply

December 24, 2019

It looks like you've placed the base resistor at the collector of the Darlington. Isn't the middle pin the base of the transistor? Was this on purpose?

CHRIS @ BCR Reply

December 27, 2019

Hi Jason,
That is incorrect, pin 1 is the Base on a TIP120 (left most pin with text facing)

JOSH Reply

January 7, 2020

Hi,
I used the same schematics except replaced the relays with a push pull solenoid actuator. The data sheet says it's requires 12VDC, 1.5A
The circuit itself works but the solenoid doesn't retract well. If I push the pin in with my finger, the magnetized coils keep them in. But without the initial push, the solenoid doesn't move.
I tried replacing the solenoid with a 5v solenoid but the same thing happens.
What should I do?

ERIC SOBERANO Reply

February 25, 2020

It was mentioned:
"This solenoid valve could easily be used with the flow meter featured in our last tutorial to create a system that only allows a certain volume of water to flow before shutting off."
Can you please let me know how would you combine the code for these tutorials to achieve this?

VICTOR OLUYOLE Reply

April 3, 2020

Good morning brother. Thank you very very much for the very clear explanation. God bless you Sir.
Sir, you said you will talk about incorporating a flow meter System. Can I please have the link to that? I am working on a Water-Cut System so I need the inclusion of a flow meter to control the Solenoid actuator after a certain volume of water has passed through.
I will be very appreciative if I can get a link to that Flowmeter project interfaced with the Solenoid valve controlled by Arduino.
Thank you very much Sir. Regards.

ADAM LUPINACCIO Reply

April 23, 2020

How could you make this ir triggered? Like a 3 pin ir reciever? Please help!

CRESENT Reply

May 6, 2020

Hi I would like to know that can I control solenoid without using a switch

TOM Reply

July 11, 2020

Hi, many thanks for the tutorial, it works great!
Is it possible to power the solenoid valve with AA or AAA batteries? Happy tu use a relay or any other required component. I would like to maximize how long the batteries will last and minimize the number of batteries needed. Thanks for your work!

CMG Reply

September 9, 2020

Hello, I'm new to arduino and electronics. I have an 12v solenoid valve as well but my power supply is only 5volts. How can i power valve with 5 volts? I thought connecting the battery to arduino vin and connecting a voltage regulator to 5v out. Then build the same system. I really liked your tutorial, pls could you help me? Thanks in advance…

ANIRUDDHA Reply

September 28, 2020

can we use arudino nano istead of aurdino uno??

ELLIS Reply

December 3, 2020

I'm wondering what I should use to power this and how I would connect it?

SOMEONES Reply

December 26, 2020

Hello have you figured out why? Cause my solenoid valve does the same thing – only hearing clicking noises but the value is not opening. I'm also supplying 9v through an Arduino.

WANA Reply

March 14, 2021

Hi. is this tutorial suitable for gas solenoid valve.? I already follow your tutorial step by step however the valve are not turned on. Do i required to connect it with the gas tank first ?
p/s : i currently done a project to store gas in a tank and use the solenoid valve to ensure the gas can be release at specified time .

DEEPAK Reply

April 23, 2021

Excellent illustration, even bare knowledge one can quickly and easily understand. Big thanks

---

KOGUL Reply

May 23, 2021

I'm trying to do this setup but using Pi Pico instead of Arduino to emit the control signal. But my solenoid doesn't work. Any help is appreciated – here is the link to my connection – https://imgur.com/gallery/EkLP4Vm

---

INDERPREET SINGH Reply

June 10, 2021

How to control 12v solenoid valve with Arduino. Instead of using breadboard can we use 5V relay

---

TAHIR Reply

June 15, 2021

First thing, great job with tutorial, very comprehensible.
I am looking to drive a solenoid that draws around 2.5Amps, wondering how that will effect the schematic?Using the Arduino Uno board, I believe Vin only supports up to 1A of current.
What would be your advise? Thanks.

WILLIAM @ BC ROBOTICS

Reply

June 15, 2021

2.5A means you would probably not want to use a breadboard either – but overall, the schematic does not need to change much. The easiest way would be to power it from the “other side” of the Arduino:

Just feed the power into the common VIN line between the Arduino and the Solenoid and it will still operate the same. One key point – powering the Arduino this way means there is no reverse polarity protection so be careful!

STATHIS

Reply

July 13, 2021

Does the Arduino Need to have a 12V supply for this to work?
I’ve been powering mine using the USB coming out of my laptop and a 9V battery but nothing ever makes it trigger or even pass any current though (I’m measuring it with a multimeter)

ROBERT POTTS

Reply

August 9, 2021

I have the solenoid working with a 9 volt battery but I cannot eliminate the usb connection and keep the solenoid working.

MARJORIE

Reply

November 3, 2021

Hi.. I tried following this tutorial using a brass 12V solenoid valve (newly purchased). I couldn’t hear ticking sound (as an indicator that it is working), but when I touched the valve, the upper portion(casing of the solenoid area) was heating up. What does it supposed to mean? Please enlighten me. Thank You.

HIMANSHU

Reply

November 9, 2021

Hi!
i have a 12V adapter that i want to use to power the solenoid…. how does that work? where does that fit into the circuit? thank you

LEA

Reply

January 18, 2022

Hi! I am trying to make this work using the tutorial and schematic, but I have only connected the arduino to the PC using the USB. None of the pictures or schematic inlclude a description of how and where to connect the 9V battery. Can someone explain or show how the battery should be connected and if the USB can be connected at the same time?

KATIE

Reply

December 2, 2022

Does this exact transistor need to be used

LACEY WEIBEL

Reply

October 1, 2023

Just one comment. I tried this tutorial and only had one problem. You are not connecting the circuit to a power supply. The Vin pin is not a power supply. In order to make this work I had to plug it into the 5V power supply on the Arduino board.

CHRIS @ BCR

Reply

October 1, 2023

VIN can provide power out, however you likely did not plug in a power source to the DC Barrel Jack on the Arduino (which is what provides 9V in this case)

SAM KNIGHT

Reply

March 12, 2024

How can I set it up so that the solenoid opens when a motion sensor has not gone off after a set amount of time.

LEAVE A REPLY

Your email address will not be published.

Comment

Your Name *

Your Email *

Website

Save my name, email, and website in this browser for the next time I comment.

Post Comment